

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 08.2

Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB.

Name: Dhinakaran. S Reg. No.: 192412353

Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
%% Input Signal
Fs = 1000; % Sampling Frequency
t = 0:1/Fs:1;
x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);
%% FIR Low-Pass Filter Design
N = 20;
Fc = 0.4;
h = fir1(N,Fc,hamming(N+1));
%% Decimation Factor
M = 2;
y_dec = upfirdn(x,h,1,M);
%% Interpolation Factor
L = 2;
y_int = upfirdn(y_dec,h,L,1);
%% Adjust Length for Comparison
if length(y_int) > length(x)
    y_int = y_int(1:length(x));
elseif length(y_int) < length(x)
    x = x(1:length(y_int));
end
t1 = (0:length(x)-1)/Fs;
t2 = (0:length(y_dec)-1)/(Fs/M);
t3 = (0:length(y_int)-1)/Fs;
figure('Name','Polyphase Decimation and Interpolation','NumberTitle','off');
subplot(4,1,1); plot(t1,x,'LineWidth',1.5); title('Original Input Signal'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(4,1,2); stem(t2,y_dec,'filled'); title('Decimated Signal (M = 2)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(4,1,3); plot(t3,y_int,'LineWidth',1.5); title('Interpolated/Reconstructed Signal (L = 2)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(4,1,4); plot(t1,x,'LineWidth',1.5); hold on; plot(t3,y_int,'--','LineWidth',1.5); title('Comparison of Original and Reconstructed Signals');
disp('POLYPHASE DECIMATION & INTERPOLATION');
fprintf('Original Signal Length = %d\n',length(x));
fprintf('Decimated Signal Length = %d\n',length(y_dec));
fprintf('Interpolated Signal Length = %d\n',length(y_int));
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
%% FIR Low-Pass Filter Design
N = 20;           % Filter Order
Fc = 0.4;         % Normalized Cutoff Frequency
h = fir1(N,Fc,hamming(N+1));
%% Decimation Factor
M = 2;
% Polyphase Decimation
y_dec = upfirdn(x,h,1,M);
%% Interpolation Factor
L = 2;
% Polyphase Interpolation
y_int = upfirdn(y_dec,h,L,1);
%% Adjust Length for Comparison
if length(y_int) > length(x)
    y_int = y_int(1:length(x));
elseif length(y_int) < length(x)
    x = x(1:length(y_int));
end
%% Time Axes
t1 = (0:length(x)-1)/Fs;
t2 = (0:length(y_dec)-1)/(Fs/M);
t3 = (0:length(y_int)-1)/Fs;
%% BEST GRAPH OUTPUT
figure('Name','Polyphase Decimation and Interpolation','NumberTitle','off');
% Original Signal
subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5)
title('Original Input Signal')
xlabel('Time (s)')
```

```

ylabel('Amplitude')
grid on
% Decimated Signal
subplot(4,1,2)
stem(t2,y_dec,'r','filled')
title('Decimated Signal (M = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Interpolated Signal
subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5)
title('Interpolated/Reconstructed Signal (L = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Comparison
subplot(4,1,4)
plot(t1,x,'b','LineWidth',1.5)
hold on
plot(t3,y_int,'r--','LineWidth',1.5)
title('Comparison of Original and Reconstructed Signals')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on
%% Display Information
disp('-----');
disp('POLYPHASE DECIMATION & INTERPOLATION');

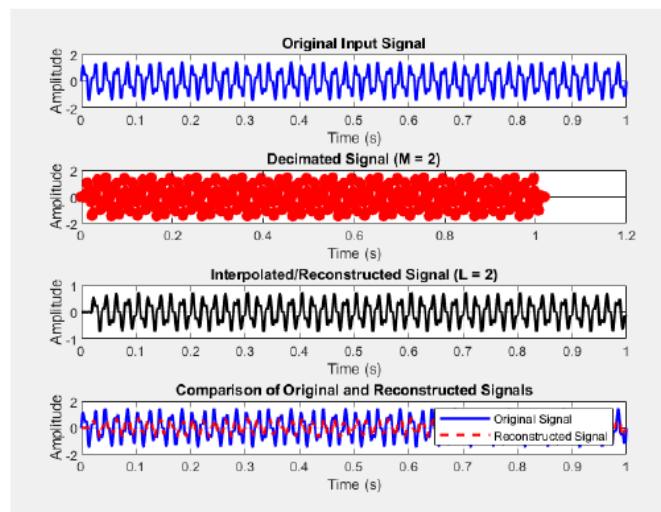
```

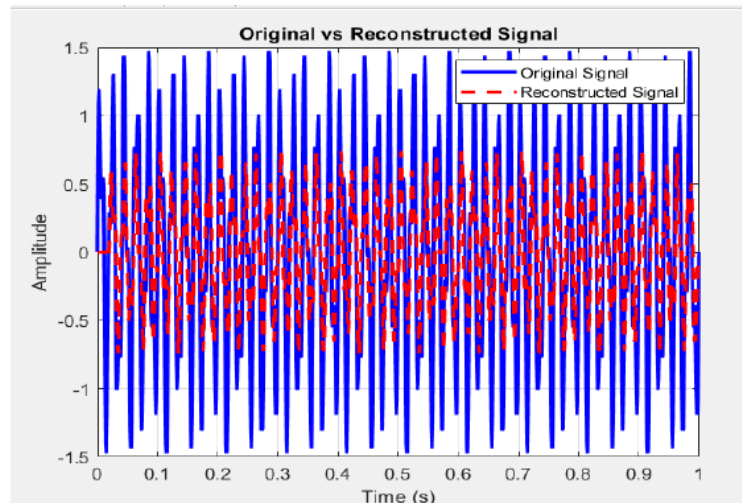
```

disp('-----');
fprintf('Original Signal Length    = %d\n',length(x));
fprintf('Decimated Signal Length    = %d\n',length(y_dec));
fprintf('Interpolated Signal Length = %d\n',length(y_int));
%% Separate Figure for Better Visualization
figure;
plot(t1,x,'b','LineWidth',2)
hold on
plot(t3,y_int,'r--','LineWidth',2)
title('Original vs Reconstructed Signal')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on

```

Output:





POLYPHASE DECIMATION & INTERPOLATION

Original Signal Length = 1001

Decimated Signal Length = 511

Interpolated Signal Length = 1001

Result :

1. The input signal was successfully decimated by a factor of 2 using a polyphase FIR decimation filter.
2. The input signal was successfully interpolated by a factor of 2 using a polyphase FIR interpolation filter.
3. The original signal was reconstructed from the decimated signal and closely matched the input signal.
4. The polyphase filter structure reduced computational complexity and efficiently performed multirate signal processing operations.

Practical Question 1

A discrete-time signal $x(n)$ is sampled at 8 kHz. Design a polyphase FIR decimation and interpolation system with a factor of 2.

Tasks:

1. Perform decimation of the input signal using a polyphase FIR decimation filter.

2. Perform interpolation of the decimated signal using a polyphase FIR interpolation filter.
3. Plot the original, decimated, and reconstructed signals.
4. Compare the reconstructed signal with the original signal and comment on the effectiveness of the polyphase structure.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Generate a sinusoidal signal of frequency 500 Hz sampled at 10 kHz. Implement a multirate signal processing system using polyphase FIR filters with a decimation factor of 2 and an interpolation factor of 2.

Tasks:

1. Design and implement the polyphase FIR decimator.
2. Design and implement the polyphase FIR interpolator.
3. Obtain and analyze the input, decimated, and interpolated output signals.
4. Evaluate the computational efficiency of the polyphase implementation and compare the reconstructed signal with the original signal.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To implement FIR-based multirate conversion using polyphase-oriented processing and compare the reconstructed signal with the original.

Student Details

Name: Dhinakaran. S

Reg. No.: 192412353

Course: ECA09 - Digital Signal Processing